

CASTING MACHINES

From molten metal to finished part — every machine family, process, defect and design rule on one poster

6,000+	years old — the oldest manufacturing process
90%	of world steel is continuously cast
±0.05 mm	finest casting tolerance (investment)
6	machine families: melt, mould, inject, spin, flow
100%	of scrap & runners re-melted — near-zero waste

THE BASICS WHAT IS CASTING?

Casting = pouring **liquid metal** into a hollow **mould** and letting it solidify into the exact shape of the cavity — the only way to make complex, hollow shapes in **one piece**.

How metal freezes: **skin-forming** alloys (iron, Al-Si) freeze from the walls inward in a clean front — easy to feed; **pasty / long-freezing** alloys (steel, brass) freeze as a mushy zone — need risers and slow, directional cooling.

The **6-step cycle**

- 1. **Melt** — metal heated to liquid in a furnace (1,400–1,700°C for iron/steel)
- 2. **Mould** — make & close mould around cavity
- 3. **Pour** — metal in through the sprue (with slag trap)
- 4. **Fill** — cavity + runners fill completely in seconds
- 5. **Solidify** — cools & shrinks ~1–2% (pattern oversized)
- 6. **Finish** — knock out, cut gates, grind flash, shot-blast

Casting is 2,000-year-old tech at scale: it dominates ~90% of steel production, ~70% of all metal parts by weight begin as a casting, and no other process can form a water-jacket engine block in one piece.

WHY CAST? WHY CAST, NOT MACHINE?

- Complex shapes & internal passages in ONE piece — no joints, no welding
- Re-entrant angles & curved internal channels no cutting tool can reach
- Any size: grams (jewellery) to 100+ tonnes (ship propellers)
- Near-net shape — very little machining waste
- Cheap per part at high volume — tooling cost is spread
- Any metal, even alloys too hard to machine
- Scrap, runners & risers re-melted — near-zero waste

Not ideal: tiny batches needing tight tolerance (machine instead); parts needing forged grain-flow strength (hammer/roll instead).

THE FAMILY TREE CASTING PROCESS FAMILIES

EXPENDABLE MOULD — destroyed to free the part

- Sand casting — most common, any size
- Shell moulding (Croning) — precision small parts
- Investment casting (lost wax) — finest precision
- Lost foam — complex one-piece castings

PERMANENT MOULD — steel die reused 100k+ times

- Gravity die · Low-pressure die
- Pressure die casting — hot & cold chamber
- Centrifugal · Continuous · Squeeze casting (chills control cooling)

THE MELTING MACHINES FURNACES — WHERE THE METAL BECOMES LIQUID

Five furnace families — choice depends on metal, batch size, cleanliness and cost.

Furnace machine	How it melts / key facts	Metals	Typical use
Cupola	Coke + scrap + limestone charged from top; air blast at tuyeres; iron melts at ~1,500°C, sinks and is tapped continuously (1–50 t/h); slag floats on top. Cheap but dirty.	Cast iron	Iron foundries: engine blocks, manhole covers, pipe fittings.
Induction (coreless)	Water-cooled copper coil around crucible; high-frequency AC eddy currents melt the charge from inside. Clean, quiet, melt-speed & temp accurate; whole furnace tilts to pour. Best for recycling scrap.	Steel, iron, Al, Cu	Modern foundries, alloy steels, recycling, kg-scale to 100+ t.
Electric arc (EAF)	Carbon electrodes strike a ~3,000°C electric arc onto scrap; roof swings open to charge; furnace tilts to tap. Rapid melt, very high power draw.	Recycled steel	Mini-mills — ~30% of world steel: rebar, sections.
Crucible	Gas/electric furnace heats a ceramic–graphite pot from outside; metal melts inside; lift-out or tilting type. Precise temperature control, cheap to run small.	Al, brass, bronze, gold/silver	Jewellery, art castings, small non-ferrous foundries.
Reverberatory	Shallow hearth; burner flame and hot gases play across the metal surface; melts bulk charges fast with large capacity. Lower melt loss than flame directly in metal.	Aluminium, Cu alloys	Bulk melting & remelting of aluminium for die casting.

MACHINE FAMILY 1 SAND CASTING SYSTEM

Expensive tooling — rapid set-up, lowest tooling cost

The machines

- Moulding boxes (flasks)** — cope (top) + drag (bottom) hold the sand
- Jolt-squeeze machine** — jolt rams sand by impact, squeeze presses it firm
- Sand slinger** — throws sand into the flask at speed for large moulds
- Automatic lines** — high-pressure, flaskless, 500+ moulds/hour
- Core shooter** — blows resin–sand into a core box, cures in seconds
- Shakeout grid** — vibrates castings free of sand

Process: pattern on matchplate → flask over it to fill + jolt-squeeze → strip → set cores → close cope on drag → pour (sprue → runner → gate) → cool → shakeout → cut gates, grind flash.

Green sand = silica + 4–8% clay + water, recycled ~95%. Cores (oil/no-bake/resin) form internal passages such as water jackets. Tolerance ±0.5–1.5 mm; any size; cheapest tooling.

Patterns: wooden (prototypes), aluminium/resin (production), wax (lost). Matchplate = pattern + runners mounted both sides — both mould halves in one machine cycle.

MACHINE FAMILY 2 DIE CASTING MACHINE

Pressure die casting — the mass production machine

Anatomy: **clamping unit** (platens + toggle, rated by **clamp force in kN**) + **die** (cover + ejector halves, H13 tool steel, water-cooled, vented) + **injection unit**.

Clamp force keeps the die shut against injection pressure: small machines ~100–500 kN, big automotive machines 5,000–60,000 kN.

HOT CHAMBER — gooseneck + plunger sit IN the molten bath; plunger forces metal up the gooseneck, through the nozzle into the die. Fast: 100–600 shots/h. For **zinc, magnesium, tin, lead alloys** (melt < 450°C).

COLD CHAMBER — metal ladled into a **shot sleeve**; plunger rams it into the die at 10–100 MPa injection pressure (keeps the melt away from the pump). For **aluminium, brass**.

Cycle: close → lock → inject at speed → hold pressure (feeds shrinkage) → cool (seconds) → open → ejector pins push part out → spray lubricant → repeat in 10–30 s.

Parts: gearbox housings, brackets, electronic & phone/laptop frames, toys, power tools. Tolerance ±0.05–0.1 mm; walls to 0.5 mm. Dies are expensive → mass production only.

MACHINE FAMILY 3 INVESTMENT CASTING LINE

Lost wax — finest precision of all casting

The machines

- Wax injection press** — water-cooled die, low pressure
- Assembly station** — patterns joined on a wax sprue ('tree')
- Shell line** — ceramic slurry dip + fluidised stucco, 6–10 coats
- Autoclave** — steam melts the wax out (collected & recycled)
- Burnout furnace** — 900–1,100°C sinters the shell, removes residue
- Pour station** — gravity, vacuum or centrifugal assist

Process: inject wax → assemble tree → build shell → dewax → fire → pour → cool → knock off shell → cut off & finish.

Best precision of all: **±0.05–0.1 mm**, near-net shape, complex internals (blade cooling holes), ANY alloy incl. Ni superalloys & single-crystal blades. Turbine blades, jewellery, golf heads, medical implants. Expensive → high-value parts.

MACHINE FAMILY 4 CENTRIFUGAL CASTING MACHINE

Spin the mould — density by rotation

Mould spins on a driven head: **horizontal axis** for long cylinders, **vertical axis** for rings & bushes. Metal poured in while rotating → centrifugal force (up to ~100×g) presses it against the mould wall and packs it dense.

True centrifugal = hollow part forms itself (pipes). Semi-centrifugal = shaped parts radiate from a central sprue (gear blanks). Centrifuging = small parts in a spinning frame.

- Dense, fine-grained structure — no shrinkage porosity
- Impurities & slag are lighter — float to the bore, machined away
- Hollow parts need NO core — the bore forms itself
- Thick clean outer wall = ideal bearing surface
- Wall thickness set by pouring speed & spin rate

Parts: spun iron/steel **pipes** (water, sewer), cylinder liners, brake drums, bushes, bearings, gear blanks; small machines for jewellery & art.

MACHINE FAMILY 5 CONTINUOUS CASTING MACHINE

One endless strand — 90% of world steel

Train: **ladle** → **tundish** (steady, slag-free flow) → **water-cooled copper mould** (oscillates; skin ~10–20 mm forms instantly) → **spray cooling** → **withdrawal & pinch rolls** → **straightener** → **flying torch** cuts lengths.

The mould is open at both ends — liquid in at the top, solid strand out at the bottom; cast speed ~1–8 m/min for billets, set by how fast the skin solidifies. Oscillation stops the skin sticking to the copper.

- Product: billets, blooms & slabs — steel, aluminium, copper
 - Replaced ingot casting: no cropped ingot tops — higher yield
 - Less energy, uniform fine grain, fully automated flow
 - Multiple strands poured from one tundish in parallel
 - ~90%+ of the world's steel is continuously cast
- Downstream: billets → rolled bar/rod; slabs → sheet & plate.

MACHINE FAMILY 6 OTHER CASTING PROCESSES

Specialists for wheels, blocks and thin walls

- Gravity die** — steel mould, gravity pour; pistons, cylinder heads
- Low-pressure die** — furnace below mould, gas pushes metal up a stalk; clean, smooth fill; alloy wheels, cylinder heads
- Lost foam** — EPS foam pattern in unbonded sand; metal vaporises foam; one-piece engine blocks
- Shell moulding (Croning)** — resin-coated sand cured on a heated pattern; camshafts, valve bodies
- Squeeze casting** — cast + forge hybrid, solidifies under pressure; dense, strong wheels & pistons
- Vacuum casting** — vacuum-assisted pour; thin sections, reactive alloys, aerospace titanium

Rule of thumb: sand when volume is low or part is huge; gravity/low-pressure die for wheels & heads; pressure die for millions of small accurate parts; investment when precision beats price.

COMPARISON PROCESS COMPARISON MATRIX

Tolerance, finish, volume, cost — pick the right process for the job.

Process	Mould	Tolerance	Finish (Ra)	Batch size	Tooling cost	Typical parts
Sand	expendable sand	±0.5–1.5 mm	6–25 µm	any	low	engine blocks, covers, pumps
Shell (Croning)	expendable shell	±0.2–0.5 mm	3–6 µm	low–med	medium	camshafts, valve bodies
Investment	expendable ceramic	±0.05–0.1 mm	1–4 µm	low	med–high	turbine blades, implants
Gravity die	permanent	±0.2–0.5 mm	2–6 µm	medium	medium	pistons, wheels
Low-pressure die	permanent	±0.2–0.4 mm	2–5 µm	med–high	high	car wheels, cylinder heads
Press die — hot	permanent	±0.05–0.1 mm	0.8–1.6 µm	very high	very high	toys, handles, housings (Zn/Mg)
Press die — cold	permanent	±0.05–0.1 mm	0.8–1.6 µm	very high	very high	Al housings, laptop/phone frames
Centrifugal	permanent (rotating)	±0.3–1.0 mm	3–12 µm	medium	med–high	pipes, liners, brake drums
Continuous	permanent (open)	±1–3 mm	12–50 µm	very high	very high	billets, slabs, rod
Lost foam	expendable foam	±0.3–0.8 mm	3–10 µm	medium	med–high	engine blocks, complex castings

QUALITY CONTROL CASTING DEFECTS — RECOGNISE, CAUSE, PREVENT

Every defect has a cause. Every cause has a fix.

Defect	What it looks like	Main cause	Prevention
Porosity (gas)	Tiny internal holes	Gas trapped in melt; damp sand; high pouring temp	Degas melt (N/C); dry moulds & cores; vent well
Shrinkage cavity	Large void at thick section	Solidification not fed; wrong feeder position	Risers over thick sections; chills to promote directional solidification
Misrun	Incomplete part, rounded edges	Metal froze before filling; too cold	Higher pour temp; bigger gates; faster pour
Cold shut	Visible seam, poor fusion	Streams met without bonding; turbulent fill	Hotter, faster pour; redesign gating; raise mould temp
Flash	Thin fin at parting line	Gap between die halves; low clamp force	Maintain dies; raise clamp force; precision machining of faces
Blowholes	Round pinholes near surface	Steam/gas from wet sand or core binder	Dry sand; vent cores; lower moisture content
Hot tearing	Crack while cooling	Contraction constrained by geometry	Fillets & radii; uniform wall thickness; collapsible cores
Slag / dross	Dirty inclusions in metal	Oxide/scum poured in from ladle	Skim, filter (ceramic filters); clean ladle; slag trap in gating
Sand inclusion	Sand spots on surface	Sand washed into cavity by fast metal	Stronger mould; calmer pour; smoother gating
Mold shift	Cope/drag halves out of line	Mould halves moved during close or pour	Locating pins; careful handling; secure clamping
Warping	Part distorts on cooling	Uneven cooling across thin/thick sections	Uniform walls; ribs; straighten after; proper cooling

DESIGN DESIGN RULES FOR CASTING

- Uniform wall thickness** — keep max ≤ 3× min to avoid hot spots
- Rounded corners** — fillets & radii at least 2–5 mm (no stress raisers)
- Draft 1–2°** on all vertical faces so the part releases from mould/die
- Risers over the thickest sections** to feed shrinkage
- No undercuts** in permanent dies — use side cores, or change process
- Machining allowance** 1.5–6 mm where accuracy is needed
- Design the parting line first** — it dictates flash, draft & cores
- Minimum hole size** ~1.5–3 mm sand / ~0.5 mm die — below that, drill

SHRINKAGE ALLOWANCE — patterns are made oversized by: zinc 0.6% · aluminium 1.3% · brass 1.5% · grey iron 1.0% · steel 2.0% · copper 1.6% · magnesium 1.4%

MATERIALS CASTING MATERIALS

Material	Why cast it	Typical parts
Grey iron	Cheap, casts easily, damps vibration	engine blocks, manhole covers, bases
Ductile iron	Stronger & tougher (nodular graphite)	crankshafts, gears, hubs
Steel	Very strong, weldable	railway parts, mining gear, valves
Aluminium	Light (2.7), corrosion-resistant, casts superbly	wheels, housings, frames
Brass (Cu–Zn)	Corrosion-resistant, attractive, easy-finish	taps, fittings, instruments
Bronze (Cu–Sn)	Low-friction bearing metal, marine-safe	propellers, bearings, bells
Zinc (Zamak)	Cheap, precise, low-melt die casting	toys, handles, die-cast trim

VOCABULARY GLOSSARY — 18 KEY TERMS

Pattern — copy of the part, oversized for shrinkage

Core — sand insert forming an internal hole

Runner — horizontal channel feeding the cavity

Riser / feeder — reservoir that feeds shrinkage

Draft — taper on walls so the part releases

Chaplet — support that holds a core in place

Flask — box holding the sand — cope top, drag bottom

Fettle — clean-up: cut gates, grind flash

Gooseneck — hot-chamber duct carrying molten metal

Mould — cavity that shapes the metal

Sprue — vertical channel metal is poured into

Gate — entry point into the cavity

Parting line — join line of the mould halves

Flash — thin metal squeezed into the joint

Chill — metal insert speeding local cooling

Shakeout — vibrating grid freeing castings from sand

Tundish — reservoir above the continuous-cast mould

Shot sleeve — cold-chamber cylinder the plunger rams

EXAM TIP

DT IGCSE 0445 — ANSWERING CASTING QUESTIONS

Remember there is a mark for the cast, 1 for 2 pouring, 1 for 2 ejector, 1 for precision, 1 for how whole, 1 for 1/2, 4 open, 5 exact, 6 trim flash.

SAND vs DIE CASTING: tooling low vs very high · rate slow vs fast · tolerance ±1 mm vs ±0.05 mm · metals any vs low-melt (hot) / Al–brass (cold) · volume any vs mass production.